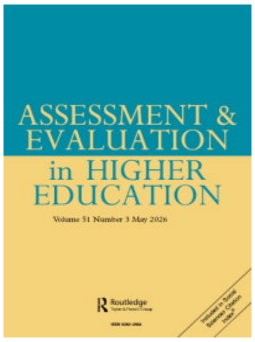

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Feedback, reflection and psychological safety: rethinking assessment for student well-being in higher education

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ABSTRACT

This study investigates how structured, timely and emotionally intelligent feedback can support both cognitive development and emotional regulation within a competition-based learning (CBL) framework in higher education. Specifically, it asks whether feedback-rich, low-stakes assessment models can reduce anxiety and enhance learner engagement in practice-based disciplines. Using longitudinal action research across four academic cycles (2021–2024), data were collected from 73 students and 19 educators *via* performance scores, surveys, learning analytics and reflective journals. Results showed consistent reductions in assessment-related anxiety, greater student motivation and improved self-regulation, with five key themes emerging. The model repositions feedback as a developmental mechanism, rather than a judgmental one, highlighting its role in fostering learner agency and wellbeing. These findings offer a scalable assessment approach that warrants further exploration across diverse disciplines and institutional settings.

KEYWORDS

Formative assessment; competition-based learning; self-regulated learning; assessment for learning; feedback literacy

Introduction

Assessment in higher education is widely acknowledged as a contributor to student stress and other negative psychological effects, particularly when characterised by high-stakes conditions that emphasise single-attempt evaluations and final grades (French, Dickerson, and Mulder 2024; Holmes 2023) While formative and authentic assessment practices have gained traction as alternatives, empirical evidence remains mixed on whether format alone meaningfully improves student well-being (Jones et al. 2021). Increasingly, attention has shifted towards the psychological experience of assessment, with feedback emerging as a potentially more significant determinant of engagement and learning than assessment type itself (Balloo et al. 2024; Kusurkar et al. 2023; Liu et al. 2025).

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Educational psychology literature consistently highlights that performance gains are not linearly correlated with increases in effort or pressure. According to the Yerkes-Dodson Law (Yerkes and Dodson 1908), there is an optimal level of arousal, including anxiety, at which performance peaks. Too little arousal can result in disengagement, while too much leads to cognitive overload and diminished outcomes. This concept has been reaffirmed in contemporary research, which suggests that moderate stress can be motivating, but excessive anxiety impairs concentration, working memory and self-belief (Cassady and Johnson 2002; Putwain and Symes 2011). Importantly, anxiety should not be eliminated but rather managed through support mechanisms that foster resilience and emotional regulation. If assessment environments in higher education embedded these supports, such as timely, low-stakes feedback and psychological safety, students might come to see academic challenge not as a threat, but as an opportunity for growth.

In elite sporting contexts, it is widely acknowledged that peak performance is not possible in environments where anxiety overwhelms focus or undermines self-belief (Gucciardi, Hanton, and Fleming 2017), the management of psychological stress has been recognised as important as skill (Pilkington et al. 2025). This recognition has led to the integration of performance and wellbeing coaches into athletes development, professionals who focus not just on physical preparation, but on mental conditioning, emotional regulation and sustained confidence. One example of this at the highest level is Angela Cullen, whose work with Sir Lewis Hamilton was widely credited with supporting his psychological resilience, focus and motivation throughout the high-pressure world of Formula One. Such coaches create a space where athletes are encouraged to view failure as informative rather than defining, using setbacks as fuel for future growth. Yu, Yang, and He (2024) suggest that well-supported athletes develop resilience and rebound more effectively from setbacks, while debriefing, and performance analytics, are not be treated as optional extras, but as core components of development. This shift reflects a broader understanding that performance anxiety is best managed not by lowering expectations, but by embedding support mechanisms that build resilience, emotional regulation and self-belief.

Despite extensive research into the effects of feedback on learning outcomes, motivation and engagement (Brandmo and Gamlem 2025; Mamoon-Al-Bashir, Kabir, and Rahman 2016), its role in supporting emotional regulation and reducing assessment-related anxiety in education remains underexplored in higher education literature. Rani (2025) highlights this as a pressing gap, calling for empirical studies that examine how feedback mechanisms can mitigate stress and promote emotional resilience in higher education. Higher education could learn from the practices in elite sports in how to manage stress and other negative psychological effects. While one-to-one performance coaching may not be feasible at scale, well-designed feedback systems could replicate many of the same benefits. When feedback is timely, personalised and low-stakes, it could offer students continuous insight, emotional reassurance and actionable guidance, serving a similar function to that of a performance coach. Such systems do more than assess performance, they help to build psychological safety, encourage self-reflection and promote resilience in the face of challenge. When embedded effectively, feedback becomes not merely evaluative but developmental, allowing students to engage with academic tasks as opportunities for growth rather than threats to be avoided, mirroring the mindset fostered in elite performance coaching.

Competition-based learning (CBL) refers to the use of structured, goal-oriented challenges to promote skill development, motivation and learner engagement through performance benchmarking. While often associated with elimination-style contests, in educational contexts CBL can be designed developmentally, centred on achieving personal bests, and iterative improvement (Martin et al. 2014). In this study, CBL was not used to rank or exclude learners, but to provide emotionally engaging, real-world scenarios that encouraged students to test, refine, and build confidence in their capabilities.

This model is also grounded in the principles of andragogy (Knowles 1970), which as Knowles argues, better suits the needs of adult learners in higher education by promoting autonomy, self-direction and relevance, qualities that traditional pedagogical approaches often overlook. Rather than treating students as passive recipients of instruction, the CBL framework positioned them as active agents who could reflect on their progress, make informed decisions and recalibrate their strategies. The structured feedback loops and low-stakes challenges offered in this model align closely with adult learning theory's emphasis on self-direction, relevancy and problem-solving. Together, CBL and andragogical principles provide a mutually reinforcing foundation for emotionally intelligent, learner-centred education.

Despite widespread interest in the impact of feedback on performance and motivation, its role in supporting emotional regulation and learner agency remains underexplored. This study responds to that gap by exploring how feedback, when embedded within a structured, iterative framework, can function not only as a cognitive scaffold but also as a tool for emotional regulation. The research questions guiding this inquiry are:

Q1: How does automated feedback support emotional regulation and the development of learner agency in higher education?

Q2: To what extent does feedback-informed practice enhance students' acquisition, retention and application of complex technical skills?

Method

Study context

This study adopted a longitudinal action research methodology to examine the affective and developmental impact of structured feedback within a CBL framework. Action research was selected for its capacity to generate practice-informed knowledge through iterative cycles of planning, implementation, evaluation and refinement (Kemmis, McTaggart, and Nixon 2014; McNiff 2016). The design supported close alignment between intervention and context, with the researcher acting reflexively as both practitioner and investigator. Feedback mechanisms were a central analytical focus, explored for their dual role in cognitive development and emotional regulation (Carless and Boud 2018).

CBL functioned as an andragogical scaffold, that is, a structured, adult-learning framework that supported students through progressively more complex challenges, designed to build competence, autonomy and emotional resilience (Knowles 1970). Learners were encouraged to reflect, self-direct and improve through feedback-rich tasks. If they struggled, they did not experience total failure but could return to a previously mastered level, maintaining motivation while reinforcing learning. This safe, incremental approach is consistent with andragogical principles that emphasise real-world relevance, self-evaluation and learner autonomy. Learners engaged in timed, performance-based tasks modelled on industry-standard assessments. Unlike traditional summative formats, CBL here emphasised iterative improvement, peer benchmarking and real-time feedback, mirroring the high-pressure but developmental ethos of competitions. The approach aimed to replicate authentic conditions while embedding formative feedback at key moments of task engagement. Feedback mechanisms were a central analytical focus, explored for their dual role in cognitive development and emotional regulation (Carless and Boud 2018).

Participants and context

The study engaged 73 students and 19 educators across 19 UK higher education institutions. Students were enrolled in built environment programmes such as architectural technology, building services engineering, quantity surveying and construction management. The activities were embedded within credit-bearing coursework, developed in accordance with national occupational standards to reflect authentic professional expectations (Gulikers, Bastiaens, and Kirschner 2004).

The integration of the intervention varied across participating institutions. At Education Institution 1 (EI1), the model was fully embedded within the assessed curriculum of a module, forming a core part of module delivery. Education Institution 8 (EI8) implemented the model partially within coursework but supplemented it with co-curricular elements and standalone summative assessment. In all other institutions, the model was delivered as a standalone, extra-curricular intervention, offered to students as an optional enhancement to their programme. Despite these differences, all participating institutions had at least one educator who acted as a local facilitator, supporting delivery, coordinating feedback and collecting contextual reflections. These facilitators played a key role in adapting the model to their institutional context while maintaining consistency in core delivery principles.

Study design and implementation

Across the four academic cycles (2021–2024), data collection followed a consistent annual structure. Each year, students engaged in a number of formative assessments and three summative assessments embedded throughout their module or offered as co-curricular experiences. After each assessment, educators gathered performance scores based on accuracy, and task completion. In addition to these scores, educators recorded observations on learner behaviours such as focus, goal setting, coping with pressure and use of feedback. Engagement data, such as login frequency, exercise completion rates, and task reattempts, were monitored monthly using learning analytics platforms and institutional VLEs. These metrics were reviewed in regular staff debriefs to adjust delivery and monitor student progress. In 2023, the study incorporated structured emotional reflection surveys that included Likert-scale items and open-ended prompts, completed at the end of each intervention. These were designed to capture affective and metacognitive responses to feedback and challenge. While a small number of students engaged across more than one year, cohorts were treated as independent samples. No continuous longitudinal tracking of individuals was undertaken. However, the consistency in intervention design and data collection methods across years supports a cumulative evaluation of the model's developmental impact.

The researcher, and industry partners, co-designed and co-delivered formative and summative assessment activities. These included challenge-based tasks with clearly defined scoring rubrics, structured reflection checkpoints and the opportunity for iterative resubmissions (i.e. multiple, repeated task attempts based on feedback, allowing students to improve performance over time), and participation was voluntary. Data collection was anonymised, ensuring alignment with general data protection regulations (GDPR) and institutional ethical protocols (Denscombe 2017).

CBL provided the mechanism for feedback to be experienced not as *post-hoc* commentary but as a dynamic, integral component of the learning process. The research was conceptually structured using Lewin (1946) action research spiral (Figure 1). Each cycle represented not only an opportunity for implementation and evaluation but also a moment for reflective synthesis, which directly informed subsequent planning. The spiral was interpreted here as both a methodological foundation and a metaphor for iterative andragogical development. An adapted version of Lewin's model was used to visualise the process (Figure 1). The revised spiral highlights a 'Reframe' link to capture how affective and performance outcomes from each cycle explicitly shaped future curriculum design. By foregrounding structured feedback as a trigger for both reflection and progression, the study framed CBL as an evolving, evidence-based andragogy. Using Lewin (1946) spiral model of action research, this study emulated how athletes are supported through structured coaching, quality feedback and iterative performance trials that build confidence under pressure without fear of failure. Structured feedback within cycles of planning, action, reflection and reframing through automated performance feedback, allowed learners to monitor progress, reflect, adapt strategies and re-engage with tasks, making feedback both immediate and actionable (Carless and Boud 2018; Nicol 2021).

Over four academic cycles (2021–2024), interventions were delivered through mainstream lessons, utilising a virtual learning environment (VLE) that provided both instructional content and automated feedback. Each academic year constituted a self-contained yet connected phase of action research, enabling insights from one cycle to inform and shape the next. Students engaged in performance-based tasks emulating real-world digital construction challenges. Feedback was delivered *via* automated scoring, benchmark comparisons and reflective tasks, with assessment structures designed to encourage retry, reflection and metacognitive growth. Educators contributed observational data and reflections after each cycle to support the model's refinement.

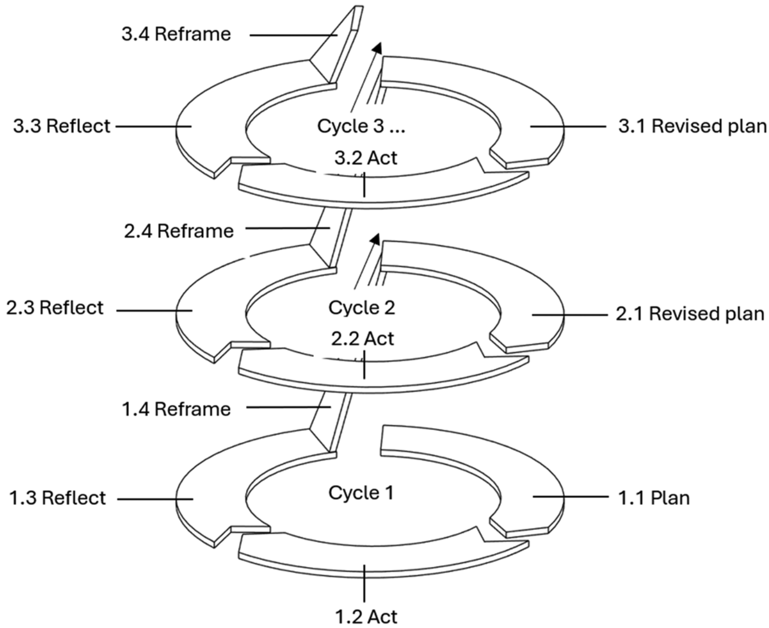


Figure 1. Adapted model of Lewin (1946) action research spiral. This visualisation emphasises the cumulative and iterative nature of the research process. The upward trajectory of the spiral symbolises the progressive refinement of the feedback-centric CBL model across time.

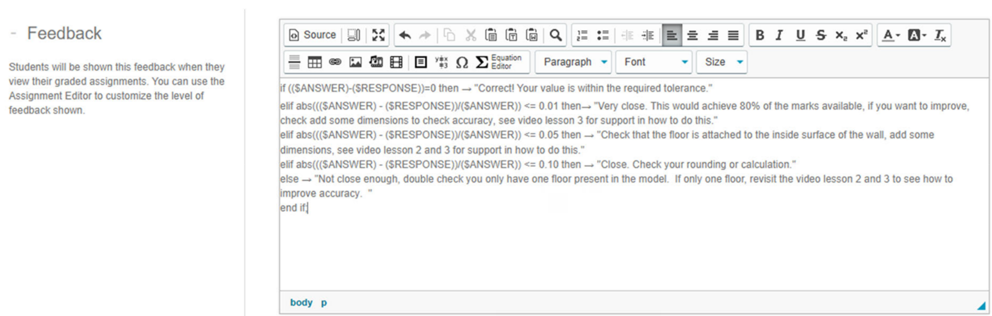


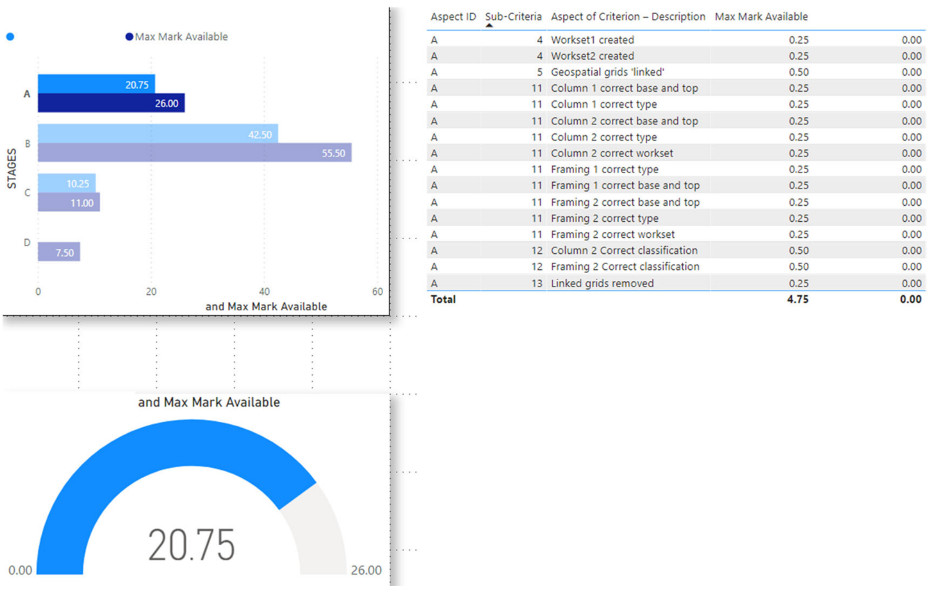
Figure 2. Example of maple script in möbius used to trigger automated, context-specific feedback.

To mirror the personalised, real-time guidance characteristic of performance coaching in sport, yet deliver it at scale, feedback was delivered automatically through Möbius using Maple scripts embedded in Blackboard (Figure 2), with responses to typical student errors, pre-programmed based on prior submissions, when students submitted a matching incorrect value, tailored guidance was provided (Figure 2 and Table 1), and the researcher could extend the feedback bank dynamically to accommodate emerging patterns, with future potential for semantic web technologies to support a scalable, national level shared error-detection framework.

Initially, visual feedback was introduced through dashboards, such as power business intelligence (Power BI), which provided colour-coded cues to help students

Table 1. Categorised types of automated feedback provided to students during competition-based tasks.

Feedback yype	Illustrative comment
Growth-oriented feedback	You're making real progress here and would pass the exercise. There's room to improve the model accuracy, but this is a solid foundation. You are showing a great level of knowledge of the software tools. If you'd like to retry and improve, add some dimensions and check the accuracy of the model. See video lesson 3 for support in how to do this.
Reflection prompt	Even if this attempt didn't fully meet the brief, your effort shows clear thinking. What would you try differently next time? How could adding dimensions improve the accuracy? See video lesson 3 for support in how to do this.
Emotionally regulating prompt	It's completely normal to feel uncertain when working with unfamiliar tools, this is a natural part of mastering something new. What matters is that you're engaging, experimenting, and taking steps forward. If you're unsure about model accuracy, revisit video lesson 3 and take the opportunity to retry. Each time you try the exercise strengthens both your knowledge and your skill.
Specific actionable guidance	To strengthen your model, try adding aligned dimensions to improve precision. Focus on key reference planes, walls, windows, and levels. Then recheck the measurement accuracy using the guide in video lesson 3. This will help ensure your model meets both technical and visual standards.



Aspect ID	Sub-Criteria	Aspect of Criterion – Description	Max Mark Available	
A	4	Workset1 created	0.25	0.00
A	4	Workset2 created	0.25	0.00
A	5	Geospatial grids 'linked'	0.50	0.00
A	11	Column 1 correct base and top	0.25	0.00
A	11	Column 1 correct type	0.25	0.00
A	11	Column 2 correct base and top	0.25	0.00
A	11	Column 2 correct type	0.25	0.00
A	11	Column 2 correct workset	0.25	0.00
A	11	Framing 1 correct type	0.25	0.00
A	11	Framing 1 correct base and top	0.25	0.00
A	11	Framing 2 correct base and top	0.25	0.00
A	11	Framing 2 correct type	0.25	0.00
A	11	Framing 2 correct workset	0.25	0.00
A	12	Column 2 Correct classification	0.50	0.00
A	12	Framing 2 Correct classification	0.50	0.00
A	13	Linked grids removed	0.25	0.00
Total			**4.75**	**0.00**

Figure 3. Power BI visualisation of student performance scores used for formative feedback.

interpret their performance progress at a glance (Figure 3). However, as students increasingly inquired about their relative standing within the cohort, the feedback system was adapted to offer more personalised insight while preserving confidentiality. A new method was developed using Google Sheets and Google Apps Script, whereby each student received a unique, access-restricted spreadsheet containing only their own scores, feedback comments and a traffic light indicator of performance, green for high, amber for average, red for below cohort average (Figure 4). These visual cues were derived from class-wide performance distribution but delivered privately, striking a balance between motivational benchmarking and safeguarding student privacy.

Aspect ID		Optimum marks	Aspect of Criterion – Description	Max Mark Available		Actual Mark
A		14.000	Structural Task A	14.000		14.000
B		28.250	Architectural Task B	28.250		28.000
C		17.750	Structural Task C	17.750		16.500
D		27.000	Architectural Task D	27.000		23.250
E		8.000	Clash Detection Task E	8.000		6.000
F		5.000	Visual Inspection Issues Task F	5.000		2.000
			Total score	100.000		89.750
						1

Figure 4. Excel visualisation of student performance scores used for formative feedback.

All individual dashboards were dynamically generated and centrally controlled *via* a master sheet accessible only to the instructor, ensuring both scalability and consistency of feedback delivery across large cohorts. This approach not only enhanced metacognitive awareness and learner self-regulation, but also mitigated anxiety often associated with public performance comparisons (Winstone et al. 2017). Although the immediate feedback system was designed to be responsive, the positional feedback component, indicating a student's relative standing within the cohort, did not update in real time. Instead, the researcher manually extracted performance data from the VLE, integrated it into a master spreadsheet, and revised feedback prompts in response to emerging patterns of error and misconception. This semi-automated workflow allowed for adaptive, data-informed guidance while preserving instructor oversight and pedagogical control. Students received immediate access to their latest scores and comments, but visibility into their relative position within the cohort, *via* the colour-coded indicator, was limited to weekly updates to safeguard against performance anxiety and over-comparison (Winstone et al. 2017).

To operationalise feedback in a way that mirrors the scaffolding used by elite sports coaches, the researcher embedded formative comments that reinforce self-belief through structured, low-stakes correction and iterative improvement. This approach reduced the emotional intensity often associated with assessment and reframed challenge as an opportunity for growth, promoting resilience, reflection and self-efficacy (Table 1).

Data collection and analysis

To examine the multifaceted impact of feedback-driven assessment on student development, this study employed a mixed-methods design spanning a 4-year period. Following the initial two cycles in 2021 and 2022, which primarily focused on performance metrics and quantitative outcomes, the research emphasis shifted towards exploring students' emotional responses. Subsequent cycles incorporated qualitative methods to better understand experiences of anxiety, confidence and engagement within the competition-based framework. Data sources were selected to capture emotional, cognitive and behavioural dimensions of learning. Annual post-intervention surveys combined Likert-scale items with open-ended questions exploring student confidence, emotional responses, motivation and perceptions of feedback utility (Winstone et al. 2017). An earlier exploratory study conducted by the author between 2018 and 2019 focused exclusively on collecting student performance scores during competitive modelling tasks. These data captured measurable outputs such as task accuracy, completion time and scoring consistency within simulated industry scenarios. While the study was not formally published, it played a critical developmental role in shaping the design of the subsequent research. Observations from this preliminary phase highlighted the need to investigate not just technical outcomes, but also emotional and cognitive responses to feedback under competitive conditions, ultimately informing the feedback-centric and reflective dimensions of the current study. In 2021, the model evolved to include structured monthly reflective evaluations. Provided with reflective logs, students were not limited in their answers and were asked:

- Task interpretation
- Did I fully understand the task?
- Did I ask sufficient questions to clarify the task?
- Was the task clear and unambiguous?
- Were there any issues with the task?
- Did the timetable clearly show my work times?
- Was the equipment working?
- What went well with your performance in the session?
- Discuss what did not go well with your performance in the session?
- What do I plan to do to improve my performance?
- What did I learn in this session?
- How do you rate your performance in this session?
- On a scale of 1-10 1 = totally unacceptable, 10 = Could not do better

These reflections marked a deliberate shift towards metacognitive engagement and emotional insight, aligning with Kolb's experiential learning cycle (1984), particularly the phases of reflective observation and active experimentation. By encouraging learners to articulate experience, evaluate outcomes and plan strategic improvement, the activity supported deeper internalisation of feedback and promoted the development of feedback literacy (Carless and Boud 2018). In 2023, the reflection process was refined into a structured, online survey format combining Likert-scale items with open-ended prompts. This allowed for more consistent data analysis across cohorts while preserving opportunities for personal reflection. The move from narrative logs to survey responses also aimed to reduce participant fatigue and increase completion rates, without sacrificing insight into learner experience.

Across all formats, the reflective components served as a key mechanism for emotional regulation, resilience-building and performance improvement, consistent with pedagogical models that view learning as iterative, student-centred and affectively grounded. In parallel, educators observed and documented students' behavioural and psychological attributes throughout the challenges, drawing on principles from performance coaching. These included: self-awareness, coping with pressure, planning and organisation, goal setting, technical performance under pressure, focus and distraction control and willingness to seek social support. Additional indicators such as health and wellbeing management, use of visualisation techniques, and realistic performance self-evaluation were also captured. These observations, while qualitative, offered valuable insights into learners' emotional development and adaptive behaviours in response to structured feedback and challenge. The integration of reflection and coaching-style monitoring enabled a richer understanding of student progress across cognitive, behavioural and affective domains.

Educator perspectives were captured through annual reflective accounts and semi-structured interviews, offering insights into student engagement patterns, feedback responsiveness and perceived emotional shifts in the classroom (Brown 2020). Additionally, anonymised learning analytics were extracted from the VLE and YouTube, including metrics such as login frequency, time-on-task, performance trajectories and resubmission rates. These behavioural data provided quantitative evidence of iterative engagement and learning progression, and were collected in full alignment with GDPR and institutional ethical protocols (Denscombe 2017). In 2023, both students and educators completed an online evaluation survey designed to measure the perceived impact of the CBL intervention on learning and emotional development. Participants responded to eight Likert-style questions on a scale of 1–5 (5 being most effective), comparing the intervention to traditional education methods across several domains:

On a scale of 1–5 (5 being most) how effective do you believe this study when compared to traditional education methods was in the following areas:

Q1: Help you/your student's grasp complex concepts more quickly?
Q2: Help you/your student's knowledge retention of complex concepts?
Q3: Make learning more interesting for you/your students?
Q4: Engage you/your students more in lessons?
Q5: Develop you/your student's critical thinking and problem solving?
Q6: Develop you/your student's knowledge and understanding of a subject to a deeper level?
Q7: Have a positive impact on your student's confidence and self-belief?
Q8: Have a negative impact on you/your student's confidence and self-belief?

The survey for this study achieved an overall response rate of 75.47%, again reflecting a high level of engagement among participants. Specifically, 25 out of the 34 students who took part in the 2023 and 2024 cycles responded, yielding a 73.53% response rate (Figure 5), while 15 of the 19 educators/mentors responded, giving a similar response rate of 78.95% (Figure 5). This high response rate from both groups aligns well with research standards suggesting that rates above 70% are generally strong and indicative of data reliability (Dillman, Smyth, and Christian 2014). Similarly, Baruch and Holtom (2008) identify response rates over 70% as robust in terms of data representativeness, helping to ensure that findings accurately reflect the target population.

While this study achieved a strong response rate, the relatively small sample size of 15 educators and 25 students may limit the generalizability of the findings. Acknowledging this limitation, the researcher applied Dillman, Smyth, and Christian (2014) 'Tailored Design Method', incorporating strategies such as personalised invitations, follow-up reminders and concise, accessible instructions to enhance participant engagement and feedback quality. This approach aimed to address potential response bias and increase the study's reliability and relevance, despite the modest sample size. Given these considerations, the response rate in this study strengthens the credibility of the survey data and provides a solid foundation for analysis.

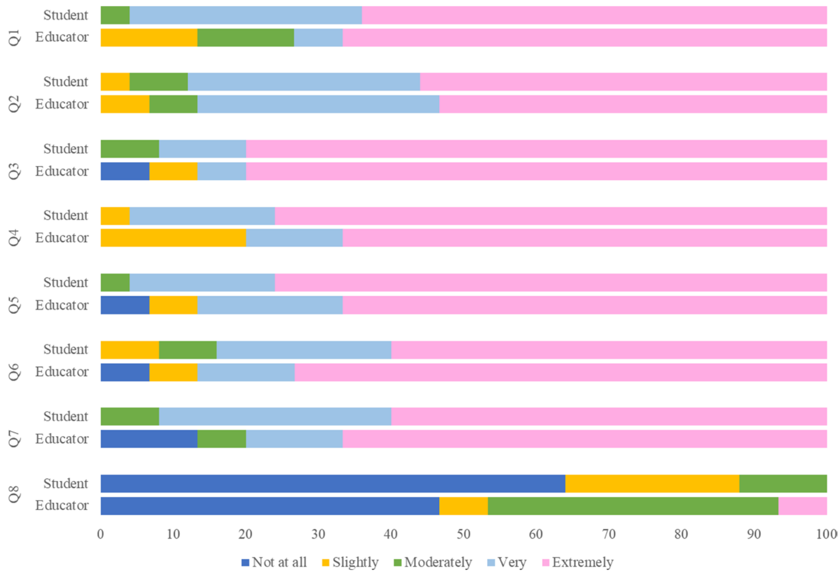


Figure 5. Descriptive results of educator and student responses to survey questions.

Analytical strategy

Thematic analysis (Braun and Clarke 2006) was used to explore qualitative data, beginning with inductive coding to identify patterns related to motivation, feedback interaction and emotional climate. A second coding cycle examined emergent themes of self-regulation, agency and resilience. To ensure trustworthiness, coding validity was enhanced through peer review and reflective memoing (Nowell et al. 2017).

Descriptive statistics were used to interpret quantitative data from surveys and analytics, revealing trends in performance, engagement and behavioural patterns over time. These findings were triangulated with qualitative data to generate a comprehensive account of how structured feedback environments influence learner development, both academically and emotionally.

Findings

The survey questions were designed to align directly with the study's thematic focus areas. For example, Q1 and Q2 (grasping complex concepts and knowledge retention) relate closely to the theme of *incremental gains* (Section 4.5). Q3 and Q4 (interest and engagement) reflect student perceptions of *emotional safety* and *motivational impact* (Sections 4.1 and 4.5), while Q5 and Q6 (critical thinking and deeper understanding) support the findings around *reflective engagement* and *feedback-driven learning* (Sections 4.2 and 4.3). Q7 and Q8 (confidence and self-belief) map directly onto themes of *emotional resilience* and *psychological safety* (Section 4.1). This alignment ensured that the survey data not only offered descriptive insight into student perceptions but also triangulated with the qualitative themes emerging from reflections and educator interviews.

The longitudinal analysis revealed that feedback, when structured, automated and embedded within a CBL framework, had a significant and positive impact on students' emotional resilience, motivational dynamics and reflective engagement with assessment. Drawing from reflective surveys, educator interviews and learning analytics over four academic cycles, five interrelated themes emerged. These findings provide evidence for the role of automated feedback not only as a cognitive aid but as an emotional scaffold that supports psychological safety and learner development (Braun and Clarke 2006; Nowell et al. 2017).

Emotional safety through predictable, non-punitive feedback

Contrary to common assumptions that assessments exacerbate stress (Posselt and Lipson 2016), participants in this study reported a notable reduction in assessment anxiety when feedback was structured, supportive and delivered immediately *via* automated systems. Rather than perceiving feedback as judgemental, students described it as an objective and actionable guide that clarified expectations and reduced emotional threat. This reframing fostered a sense of psychological safety and control, with many students describing the process as motivating and confidence-building. One participant noted, 'Seeing improvements over the different milestones increased my confidence in my skill, abilities, and knowledge', while another reflected, 'It made me realise that I am a talented engineer, rather than just a competent one'. Even when mistakes were made, students interpreted these as part of the learning process, with comments such as, 'You aren't going to do well every time… everyone will make mistakes' and 'If you expect to fail and learn from it, then it cannot be considered a negative'. These responses suggest a cognitive reappraisal of failure, aligned with resilience-building pedagogies. Rather than diminishing self-belief, the competitive format, when paired with timely and emotionally attuned feedback, was seen as an opportunity for growth. Educators similarly observed greater openness to critique and a willingness to re-engage with tasks, suggesting that a reduced emotional burden can support more adaptive learning behaviours (Rowe and Fitness 2018).

A notable divergence emerged between how educators and students perceived the emotional and pedagogical impact of the study. Figure 5 illustrates student and educator responses to the feedback and assessment survey, where student participants reported largely positive experiences, frequently selecting higher scale values for items such as increased motivation, deeper learning, and critical thinking development. Notably, no students selected 'Not at all' for any positively framed question, except in response to whether the study negatively affected them, which in itself reflects a favourable perception.

In contrast, educators' responses in Figure 5 were markedly more reserved. Some educators selected 'Slightly' for nearly all questions, including those related to engagement (Q4), critical thinking development (Q5), deeper learning (Q6), and overall positive impact on students (Q7). In several instances (Q3, Q5–Q7), educators selected 'Not at all,' highlighting a stark contrast in perception to the more positive student responses. Students tended to cluster their responses at the more affirmative end of the scale, educators remained cautious, with no selections at the upper end and a general hesitance to endorse strong developmental impact.

This disparity suggests a perceptual gap regarding the affective and cognitive benefits of the model. While students appeared to internalise the structured feedback as motivating, empowering, and developmentally valuable, educators were more inclined to interpret the competitive format through a lens of performance pressure and emotional risk. This contrast underscores the importance of triangulating learner analytics, qualitative feedback and educator reflection to fully understand how innovative pedagogical models are experienced and interpreted across stakeholder groups.

One educator remarked that 'Competition is also very much associated with pressure to perform', arguing that self-doubt is inevitable and that mental preparedness is as essential as technical ability. This aligns with findings in performance psychology, where athletes often require both cognitive and emotional coaching to perform under stress (Mesagno et al. 2021). However, despite these concerns, no students in this study explicitly reported feelings of anxiety during tasks. This absence of self-reporting, however, should not be taken as evidence that performance pressure was absent.

As Mesagno et al. (2021) caution, individuals may suppress or lack awareness of stress while still exhibiting behavioural signs, such as hesitation, errors in judgment or avoidance behaviours. For instance, one student opened an incorrect file at a crucial assessment point, reflecting a possible manifestation of underlying performance anxiety. These subtle cues suggest that traditional self-reporting tools may be insufficient for identifying emotional strain.

Reflection as a mechanism for emotional regulation

Structured reflection activities served as a critical link between feedback and emotional resilience, offering students a space to cognitively reappraise their performance and plan adaptive responses. Drawing on Kolb (1984) experiential learning cycle, students were prompted to describe what went well, what did not go well, what they had learned and how they would improve. This process supported all four stages of Kolb's model: concrete experience (task completion), reflective observation (evaluating performance), abstract conceptualisation (identifying patterns or strategies), and active experimentation (planning next steps).

Evidence from student reflections substantiates this developmental trajectory. For example, one student wrote:

'Overall I was unhappy with my performance as I felt there were a lot of small mistakes that I was making… To improve I will work on all of the things that I got wrong… and repeat the exercise.' (Student 7)

Another reflected:

'My classifications worked this week which was my problem I stated last week, so I now found a new way to work it… I now feel much more confident.' (Student 1)

Such comments demonstrate a clear progression from initial disappointment to strategic recalibration, reinforcing the emotional and cognitive value of reflective tasks.

Educators leveraged these insights to better understand individual learner needs and co-develop targeted support resources. This bidirectional process strengthened the feedback culture, moving it from a transactional exchange to a relational and emotionally attuned dialogue (Winstone et al. 2017). For example, educator reflections often noted how students who initially underperformed later demonstrated increased focus and motivation following structured feedback and self-reflection.

This iterative cycle of feedback, reflection, and adaptation also aligned with the "evaluate and reframe" phase of Lewin (1946) action research spiral. Student input and performance trends directly influenced instructional design. For instance, YouTube analytics revealed where students dropped off during video content, prompting a revision and shortening of instructional videos, which in turn increased student engagement in the following academic year.

Together, these findings underscore the central role of structured reflection – not just as a learner development tool, but as a mechanism for continuous pedagogical refinement and emotional scaffolding.

Redefining educator roles and feedback identity

As students became more feedback-literate, educators reported a parallel shift in their roles, from assessors to learning coaches. Access to detailed learner analytics and reflective narratives enabled educators to deliver nuanced, timely, and supportive interventions.

This repositioning fostered mutual trust and shared responsibility for learning, contributing to greater educator satisfaction and student engagement. The evolution of educator identity supported the development of a sustainable feedback culture rooted in dialogue, partnership and responsiveness (Brown 2020).

Equitable engagement through personalised feedback loops

Automated and iterative feedback cycles provided students with the flexibility to progress at their own pace, without fear of penalty. This approach was particularly empowering for students who historically underperformed in traditional, high-stakes contexts. By decoupling performance from public comparison and aligning feedback with individual effort and growth, the CBL model created inclusive conditions for participation. Learners were able to self-regulate, request support and engage without the stigma of failure. These outcomes reinforce calls for feedback environments that are developmentally sensitive and emotionally sustainable (Evans 2013; Nicol 2021).

Feedback-driven motivation and incremental gains

A key finding of the study was that students made consistent, measurable progress through marginal gains across multiple areas of skill development. The structured, feedback-rich environment, particularly the integration of visual dashboards and automated scoring, enabled learners to engage in self-referenced benchmarking, shifting their perception of assessment from judgment towards developmental growth. While competition is commonly associated with direct comparison against peers, often implying a zero-sum structure where one student's success limits another's, the model used here reframed competition as an intrapersonal challenge. Rather than competing against others, many students reported being motivated to 'beat my last score' or 'track my own growth', reflecting a mastery-oriented mindset more aligned with personal best goals than with normative ranking. This distinction helped to preserve emotional safety while maintaining high levels of engagement and self-regulation. These micro-improvements were not only evident in student reflections but corroborated by learning analytics, which indicated an increase in cumulative performance among students. While multiple institutions took part, only education institutions 1 and 8 (EI1 and EI8) took part every cycle (Figure 6).

This pattern aligns with the principle of marginal gains, popularised by Brailsford's work with Team Sky (Smith 2018), wherein consistent 1% improvements across many dimensions compound to produce substantial overall gains. In the context of this study, feedback was not positioned as a final evaluative statement, but as a tool for continuous optimisation, enabling students to incrementally refine technical workflows, deepen conceptual understanding and enhance strategic thinking. The impact was particularly pronounced among students with lower baseline performance, for whom the visibility of small improvements fostered psychological safety and encouraged sustained engagement.

Students' qualitative feedback reinforced this developmental trajectory. Rather than reflecting anxiety or pressure, the comments revealed a sense of confidence, enthusiasm and emotional investment in learning. Several students described the process as transformational, with remarks such as, 'By the end, my skill level was above some in the year above me who didn't take part", and "I only realised how many complex tools I understood when I compared myself to others'. Others linked incremental improvement to a deeper emotional engagement, stating, 'This was the best thing I did in four years of university", and "I have never felt a higher desire to learn'. These reflections illustrate how feedback, when timely, personalised, and actionable, can act as both a cognitive scaffold and an affective support mechanism, nurturing growth rather than anxiety.

In this context, the model becomes more than an andragogical strategy, it is a psychologically attuned framework for assessment design. By positioning feedback as iterative, non-punitive and student-facing, the model creates inclusive conditions for academic growth, particularly for those often marginalised by traditional,

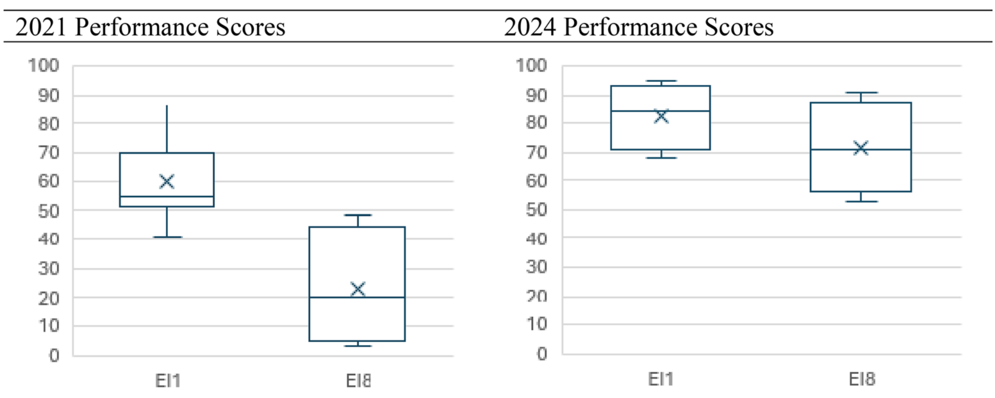


Figure 6. 2021 And 2024 UK national final performance scores highlighting the improved student outcome results.

high-stakes approaches. It also responds to calls for emotionally intelligent assessment (Hughes et al. 2018), demonstrating how continuous, structured feedback can enhance not only performance but also agency, motivation and emotional resilience.

Discussion

This study explored the role of feedback within a CBL framework as both a cognitive scaffold and a mechanism for emotional support in vocational higher education. Through a longitudinal action research design spanning 4 years, the findings suggest that when feedback is structured, timely, and framed developmentally, it can positively influence motivation, metacognition and emotional resilience. These outcomes contribute to a broader reconceptualisation of assessment as a process that is inherently psychological as well as andragogical.

Feedback as a catalyst for motivation and emotional security

One of the most compelling findings was that reports of assessment-related anxiety consistently decreased, despite the use of competitive structures. While competition is often associated with heightened stress or performance pressure (Putwain and Symes 2011), and one could assume that introducing competition into an already stressful occasion may increase anxiety levels, the carefully scaffolded and emotionally supportive design of this model, combined with instant feedback mechanisms, automated scoring, visual dashboards and transparent marking criteria, appeared to mitigate those risks. By further framing competition as a developmental tool rather than a ranking mechanism, and by embedding frequent, low-stakes opportunities for feedback and self-improvement, the model fostered psychological safety rather than threat. These features shifted assessment from a judgment to an iterative learning opportunity. In doing so, the model satisfied key psychological needs for competence and autonomy, in alignment with self-determination theory (Deci and Ryan 2000b). Ultimately, students reported feeling more in control, more confident and more emotionally regulated as they progressed, a finding that challenges the assumption that competition is inherently anxiety-inducing.

By positioning feedback as a developmental resource rather than a terminal evaluation, students were encouraged to adopt mastery-oriented goals. This was especially evident among previously disengaged learners, who demonstrated renewed confidence and sustained participation. These results echo recent scholarship advocating for feedback environments that are emotionally attuned, fair, and responsive to individual progress. Echoing Daloz (1986) assertion that challenge without support leads to retreat, the findings suggest that timely, emotionally intelligent feedback, can foster resilience and psychological safety rather than amplify stress, transforming potential performance pressure into a catalyst for growth and development.

The study found that automated feedback can serve a dual role, as both an instructional guide and a mechanism for emotional regulation. Educators can adopt coach-like approaches, using feedback to identify behavioural indicators of stress and respond with motivational prompts, structured preparation routines, and affirming language. Such feedback not only mitigates pressure but also promotes resilience. Many students reflected that the challenge of competition helped them develop adaptability and persistence, key traits often fostered by performance coaches in high-stakes environments (Daloz 1986). This reconceptualisation of feedback as a scalable and emotionally attuned support tool, offers a promising path towards reconciling educator concerns with students' lived experiences.

Reflection as a mediator between feedback and learning

While the immediacy and transparency of feedback were essential, it was the integration of structured reflection into lessons and formative assessments that provided the opportunity to transform feedback into meaningful learning. Through guided journaling, debrief discussions and self-assessment activities, students engaged in a dialogic feedback process, one that supported not just the reception of information but its interpretation, questioning and application (Carless and Boud 2018). These reflective practices fostered feedback literacy, with students increasingly able to interpret both quantitative and qualitative input and use it to recalibrate their approach.

This recursive engagement resonates with Schön's (2017) concept of the reflective practitioner and illustrates how reflection acts not merely as a supplement to feedback but as a mechanism for developing emotional regulation, resilience and metacognitive awareness. In this context, feedback was not simply delivered, it was interpreted, questioned and incorporated into a personalised narrative of growth. Informed by self-determination theory (Deci and Ryan 2000b) and experiential learning (Elliott and Dweck 1988), the model encouraged learners to interpret feedback, retry without penalty, and view progress as mastery rather than judgement.

Evolving educator roles within feedback-centric assessment

The implementation of CBL also redefined the educator's role. Moving beyond the role of assessor, educators described themselves increasingly as coaches and facilitators of learning. Drawing on performance analytics and reflective student narratives, they were able to tailor feedback, pre-empt disengagement and respond empathetically to learner needs. This shift aligns with calls for a relational model of feedback, where educators co-construct learning experiences with students, supporting emotional as well as academic development. Educator satisfaction and andragogical responsiveness also improved, suggesting that feedback-rich environments benefit both learners and teachers through enhanced relational trust and shared responsibility.

Reframing assessment culture

The broader implication of these findings is a demonstrable shift in assessment culture. Within the CBL framework, assessment became a cyclical, student-centred process rather than a singular evaluative event. Students were not ranked but supported to engage in intra-personal benchmarking and progress tracking. This developmental orientation supports arguments for emotionally intelligent, scalable feedback systems that promote well-being without diluting academic standards. In vocational and digitally mediated contexts, where skill development and iteration are paramount, the CBL model offers a viable alternative to traditional assessment designs. By embedding opportunities for revision, formative dialogue and emotional support, feedback can become the connective tissue that links performance to growth in meaningful and sustainable ways. Together, these findings affirm that when feedback is designed as a central andragogical mechanism, rather than an adjunct to grading, it can enhance both learning outcomes and emotional well-being. The CBL framework illustrates how assessment reform, grounded in feedback-rich environments, can drive student engagement, resilience and equity in practice.

Rethinking assessment: from measurement to development

Contemporary discourse increasingly recognises that the mission of higher education extends beyond the transmission of technical knowledge. Universities are expected to cultivate critical thinking, reflective judgement and the capacity for independent learning in students (Andreucci-Annunziata et al. 2023; Dwyer 2023; Mohammadi Zenouzagh, Admiraal, and Saab 2025). However, conventional assessment models, often summative, high-stakes and output-driven, are poorly aligned with these goals (Williams 2024). Such models seldom support the complex, iterative competencies required in vocational and applied disciplines and are frequently associated with heightened student anxiety and diminished motivation (French, Dickerson, and Mulder 2024). Despite a growing emphasis on student-centred education, assessment systems remain largely static. Rigid grading structures and retrospective evaluations persist, limiting opportunities for students to reflect, revise and engage with feedback in meaningful ways. This rigidity presents particular challenges in digitally enabled, practice-based domains that demand agility, experimentation and sustained engagement (Crisp 2012; Carless and Boud 2018).

While innovations such as portfolios, peer feedback and self-assessment have shown promise, their impact is often constrained by insufficient scaffolding, limited feedback literacy and institutional barriers to scalability (Sadler 2010; Evans 2013; Ajjawi and Boud 2017). This study positions CBL as a scalable, psychologically attuned assessment model that provides the mechanism for supportive feedback and reflection within authentic challenges. Mirroring elite sports, rather than treating feedback as an evaluative add-on, it integrates it into the core of learning activity, encouraging students to iterate, compare and reframe performance through structured opportunities for challenge and retry. The model fosters learner agency, emotional resilience and academic growth, particularly when situated in pragmatic contexts aligned with professional practice (Gulikers, Bastiaens, and Kirschner 2004; Van den Berg and Streumer 2011; Nicol 2021).

In this framework, assessment becomes not an endpoint but an active process of feedback-informed development. By repositioning educators as coaches and learners as reflective practitioners, CBL supports both rigour and emotional safety. This approach offers a sustainable model for high-quality, inclusive assessment, responsive to both andragogical ideals and the emotional realities of contemporary higher education. By recognising educators as coaches and positioning learners as reflective practitioners, CBL supports both academic rigour and emotional safety. This coaching paradigm aligns with definitions in the coaching-in-education literature (Heinrich et al. 2021), where educators are explicitly identified as coaches who guide student learning through reflective dialogue, goal-setting and performance feedback. Studies show that faculty acting as coaches, using check-ins, modelling, and structured reflection, enhance teaching quality and student engagement. Positioned this way, CBL offers a sustainable, scalable model: coaches (educators) embed support mechanisms directly within activities, aligning with both andragogical ideals and the emotional realities of contemporary higher education.

Reframing assessment: emotional wellness and student engagement

The emotional consequences of assessment are increasingly recognised as integral to understanding student engagement, persistence and performance in higher education. Far from being neutral measures of knowledge, assessment practices shape learners' cognitive strategies, emotional states and self-concept (Rowe and Fitness 2018). When assessments are high-stakes, opaque or offer limited feedback, they often induce anxiety, fear of failure and diminished academic confidence, especially among students who adopt maladaptive performance beliefs or perfectionist tendencies (Deci and Ryan 2000a; Winstone et al. 2017). Findings from sport psychology underscore this dynamic. Mesagno et al. (2021) demonstrated that athletes who endorsed rigid success imperatives (e.g. 'I must not fail') experienced heightened somatic anxiety (i.e. physical symptoms of stress such as increased heart rate or muscle tension), and were more likely to 'choke' under pressure. In educational contexts, similar irrational beliefs about assessment, such as equating success with self-worth, can trigger a comparable anxiety-performance loop, narrowing attentional focus and disrupting adaptive learning behaviours. These parallels point to a broader need for emotionally intelligent assessment design, where feedback and task structure buffer psychological threat and foster resilience. Crucially, assessment need not be inherently harmful. Perceptions of transparency, fairness and developmental support are consistently associated with improved affective responses (Daloz 1986; Boud and Molloy 2013; Nicol 2021). When feedback is timely, personalised and embedded within low-stakes, formative structures, it enables learners to interpret setbacks not as failures, but as opportunities for 'growth'. Such environments are more likely to promote emotional regulation, intrinsic motivation and sustained engagement.

Within this study, emotional wellness was not conceptualised as the complete absence of anxiety or stress, but as a dynamic interplay of autonomy, competence and perceived progress, core components of self-determination theory (Deci and Ryan 2000b). This view aligns with the Yerkes–Dodson law (1908), which suggests that moderate levels of stress (arousal) can enhance performance, whereas both too little and too much can impair it, while Cassady and Johnson (2002) suggest that the relationship between anxiety and performance is curvilinear, with moderate levels associated with optimal performance. The CBL model explicitly targeted these conditions by embedding immediate, layered feedback and structured opportunities for retry within assessment activities. Students described the environment as 'safe', 'motivating' and 'encouraging', particularly valuing the transparency of scoring and the non-punitive framing of performance. These findings highlight how assessment structures that normalise high-frequency, high-precision and performance-enhancing feedback, can reduce cognitive threat, including performance anxiety, promote reflective learning and reframe assessment as a developmental experience. It challenges the assumption that rigour must come at the expense of well-being and offers a viable alternative, one in which student development is supported both cognitively and affectively through purposeful feedback design.

Feedback as emotional and cognitive scaffolding

Within contemporary andragogical literature, feedback is no longer conceived solely as the transmission of corrective information, but rather as a dialogic and developmental process that supports both cognitive advancement and emotional regulation (Carless and Boud 2018). Effective feedback is characterised by its timeliness, clarity and potential to prompt student action, but crucially, it must be embedded in structures that allow learners to interpret and apply it meaningfully (Nicol 2021). Without such scaffolding, especially in high-stakes or summative contexts, feedback may provoke confusion, disengagement or diminished self-efficacy (Evans 2013; Rowe and Fitness 2018).

In this study, feedback is positioned as central to both the emotional and intellectual dimensions of the learning experience. Drawing on self-determination theory, which identifies autonomy, competence and relatedness as key psychological needs that foster intrinsic motivation and well-being (Deci and Ryan 2000b), the CBL model was designed to provide layered feedback, numerical scores, comparative benchmarks and personalised commentary, within a low-stakes, iterative framework. These feedback loops were deliberately organised around structured reflection points, allowing students to revisit challenges, adjust strategies and develop insight into their learning trajectory over time. By integrating automated performance tracking, reflective journaling and relational educator feedback, the assessment process became both emotionally attuned and cognitively expansive. Students were not only supported in identifying areas for improvement but were encouraged to understand how and why they improved. This multidimensional feedback approach fostered self-regulation, confidence and sustained motivation, particularly among students who had previously reported disengagement or anxiety in more traditional assessment settings.

Conclusion and implications for practice and research

This study demonstrates that feedback, when embedded within an emotionally supportive and reflectively structured assessment model, can act as a transformative force for both learner development and well-being. The longitudinal implementation of CBL reframed assessment from a summative judgment into a student-led journey characterised by autonomy, iterative progress and psychological safety. Within the competitive framing, feedback became a driver of engagement by encouraging benchmarking, goal-setting and iterative performance refinement (Reeve and Deci 1996).

Andragogical implications

For educators, the findings underscore the importance of designing feedback systems that are transparent, timely and actionable. Embedding structured reflection and enabling multiple opportunities for revision promotes student engagement, emotional resilience and feedback literacy. Crucially, the role of the educator evolves to support these processes, offering personalised guidance informed by longitudinal data on learner performance.

Institutional implications

At the institutional level, CBL demonstrates the potential for scalable feedback-rich environments that maintain academic integrity while supporting inclusive and emotionally sustainable practices. Particularly in practice-based and vocational disciplines, where iterative learning is essential, CBL provides a blueprint for aligning formative assessment with professional competencies and learner needs.

Directions for future research

While this study focused on pragmatic built environment education, the underlying principles of structured feedback, reflection and low-stakes iteration are broadly applicable. Future research should explore the transferability of this model across academic disciplines, delivery modes and institutional settings. Additional work is needed to investigate how feedback-centric models interact with equity, assessment literacy and learner identity, particularly among students from historically underrepresented groups.

While this study did not formally deploy validated psychometric instruments, the behavioural and reflective data were developed in consultation with trained psychologists and performance and wellbeing coaches, who helped shape the indicators used to assess emotional resilience, self-regulation and coping strategies. These expert-informed measures provided valuable insight into affective development in context. However, future research should incorporate standardised psychometric instruments, such as the Warwick-Edinburgh Mental Wellbeing Scale, to strengthen the reliability and generalisability of findings related to emotional impact.

Limitations

While the study demonstrates clear value within applied digital construction programmes, several limitations merit consideration.

First, the context-specific nature of the research, embedded in vocational disciplines with performance-based assessment norms, limits direct generalisation to more abstract, essay-based or discursive domains. Although core principles of feedback literacy, emotional safety and structured reflection hold relevance across disciplines (Carless and Boud 2018; Nicol 2021), the mechanisms of implementation may require adaptation to align with the epistemological norms of other fields.

Second, the voluntary nature of participation in surveys and reflective tasks introduces the possibility of self-selection bias. Students predisposed to metacognitive engagement or feedback-seeking behaviours may be overrepresented, potentially skewing perceptions of effectiveness. More disengaged or feedback-averse learners, those arguably most vulnerable to assessment-related stress, may have remained under-sampled, limiting insight into the full spectrum of student experience (Winstone and Nash 2016).

Third, the study relied on student self-reporting and behavioural analytics rather than validated psychometric instruments to measure emotional well-being. While this provided rich, real-time insights into affective experiences, future studies would benefit from incorporating established tools, such as the Warwick-Edinburgh Mental Wellbeing Scale (Tennant et al. 2007), to enhance measurement validity and enable comparison across contexts.

Fourth, while participants were drawn from 19 institutions and institutions took part in multiple cycles, only two institutions participated in every cycle. We acknowledge that this variability limits longitudinal consistency. Moreover, the absence of a control group means that causality cannot be definitively attributed to the intervention, although triangulated qualitative and behavioural data help mitigate this limitation.

Finally, the dual role of the researcher as both practitioner and investigator, while enriching the study with contextual authenticity and ecological validity, carries the risk of interpretive bias. This is a recognised tension in action research and practitioner inquiry (Herr and Anderson 2005; McNiff 2016). To mitigate this, future studies could incorporate independent evaluators or collaborative inquiry teams to strengthen analytical distance and confirmability.

Despite these limitations, the study offers an empirically grounded, theoretically informed, and pragmatically actionable model of assessment that integrates feedback, reflection and emotional development. It underscores the imperative to move beyond static, high-stakes models of judgement towards assessment cultures that are dynamic, human-centred and andragogically aligned with learners' psychological and developmental needs.

Disclosure statement

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